

Install Apache2 using the command given below:

```
$ sudo apt-get install apache2
```

Install HTTP Server nginx, php and other required software packages, all at once, as:

```
$ sudo apt-get install nginx openssl  
ssl-cert php5-cgi php5-sqlite php5-gd  
php5-common php5-cgi sqlite3 php-pear  
php-apc curl liblwp libtool curl  
libcurl4-openssl-dev php-xml-parser php5  
php5-dev php5-gd php5-fpm memcached  
php5-memcache varnish
```

Double-check for php5-curl. It should not be installed, so remove it by giving the following command:

```
$ sudo apt-get --purge remove php5-curl
```

Now, when you access the server, your browser might not allow you, considering it unsafe. To override that, create an SSL certificate.

Create self-signed SSL certificates for two years as given below:

```
$ sudo openssl req -x509 -days  
730 -nodes -out /etc/nginx/cert.pem  
-keyout /etc/nginx/cert.key  
$ sudo chmod 600 /etc/nginx/cert.pem  
$ sudo chmod 600 /etc/nginx/cert.key
```

Now, we need to configure Nginx Web server using the following steps:

1. Open the default file at /etc/nginx/sites-available/ location using the command:

```
$ sudo nano /etc/nginx/sites-available/  
default
```

2. Replace the content in the default file with the content provided in the default.txt file included in this month's EFY DVD. (Simply copy and paste the content in the nano editor.)

3. Replace RaspberryPi-IP in the file with the local IP of your Raspi. It should match the one provided to create the SSL certificate.

We now need to make some changes to the php and we will be ready to install ownCloud7. For that, change the default upload limit in php. To do so, open php.ini with the following command:

```
$ sudo nano /etc/php5/fpm/php.ini
```

Search for upload_max_filesize and post_max_size, and update these as given below:

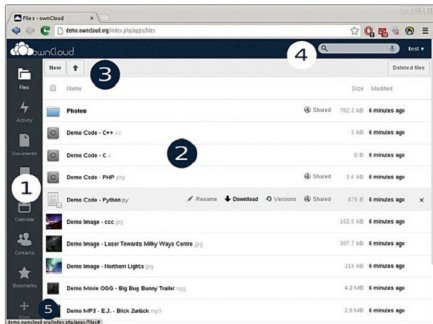


Fig. 3: Typical ownCloud window after installation

```
upload_max_filesize = 1000M  
post_max_size = 1000M
```

Next, open www.conf file using the command:

```
$ sudo nano /etc/php5/fpm/pool.d/  
www.conf
```

Replace listen = /var/run/php5-fpm.sock with listen = 127.0.0.1:9000.

Last, open dphys-swapfile with the command:

```
$ sudo nano /etc/dphys-swapfile  
Replace CONF_SWAPSIZE=100  
with CONF_SWAPSIZE=512.
```

Restart the Web server by typing sudo /etc/init.d/php5-fpm restart and php by sudo /etc/init.d/nginx restart in the terminal.

Install ownCloud by following the steps given below:

1. Create a new directory ownCloud in www folder using the command:

```
$ sudo mkdir -p /var/www/owncloud
```

2. Download ownCloud using:

```
$ sudo wget https://download.owncloud.  
org/community/owncloud-7.0.1.tar.bz2
```

If wget command does not work, you can directly download it from ownCloud's official website. If you are downloading using wget command, make sure you download the latest one.

Extract downloaded tarball:

```
$ sudo tar xvf owncloud-7.0.1.tar.bz2
```

Move the extracted folder to ownCloud directory that you created in the first step. Give permission to the user and remove the tarball using the following commands:

```
$ sudo mv owncloud /var/www/  
$ sudo chown -R www-data:www-data /  
var/www  
$ rm -rf owncloud owncloud-7.0.1.tar.bz2
```

Setting up an admin account.

You need to provide the path of the data folder where you will put all your files, which can be any folder on the SD card. If you have an NAS drive set-up, which is connected to Raspi, you can give the path of this drive. ownCloud will not work without a data directory set-up for the www-data user. It should be readable and writable by user www-data (the user we created earlier). It should also be readable for other users.

(Note. Setting up permission for the NAS drive is out of scope here. You can refer to various articles available on the Internet.)

Now, login to the IP address of Raspi as given below (where, x is the actual or local IP number in your set-up) from the browser of any computer with an Internet connection: